Distributed Multimodal Extreme Weather Risk Prediction Using Hadoop and Apache Spark with a NoSQL-Based Climate Analytics Dashboard

Akhil Sebastian
BL.SC.P2DSC25002
Department of Data Science
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham
India – 560035
bl.sc.p2dsc25002@bl.students.amrita.edu

Allwin Suresh
BL.SC.P2DSC25005
Department of Data Science
Amrita School of Computing, Bengaluru
Amrita Vishwa Vidyapeetham
India – 560035
bl.sc.p2dsc25005@bl.students.amrita.edu

I. Abstract

***Abstract*—The increasing volume of climate and environmental data has created the need for scalable big data systems capable of processing and analyzing large-scale weather information efficiently. Extreme rainfall events during the Indian monsoon season often result in floods, agricultural loss, and infrastructure damage, making rainfall risk prediction an important research area. This project presents a distributed multimodal extreme weather risk prediction framework using Hadoop, Apache Spark, MongoDB, machine learning, and an interactive Streamlit dashboard.**

The proposed system integrates multiple environmental datasets including Indian rainfall records, GPM satellite precipitation data, ERA5 climate reanalysis data, and atmospheric weather parameters such as temperature, humidity, and wind speed. Hadoop Distributed File System (HDFS) is used for distributed storage, while Apache Spark performs scalable ETL processing, data aggregation, feature engineering, and machine learning operations. The framework specifically focuses on monsoon months (June–September) to improve rainfall prediction relevance and reduce seasonal noise.

Machine learning models including Logistic Regression, Decision Tree, Gradient Boosted Trees (GBT), and Random Forest are trained using Spark MLlib for extreme rainfall prediction. Experimental results show that the Random Forest model achieves the best predictive performance. Prediction outputs are stored in MongoDB for scalable NoSQL analytics and integrated into a Streamlit dashboard that provides interactive visualizations, geospatial rainfall risk maps, hotspot detection, temporal trend analysis, and real-time climate insights.

The proposed framework demonstrates the effectiveness of combining distributed big data technologies and machine learning for scalable climate analytics and rainfall risk prediction systems.

***Index Terms*—Big Data, Apache Spark, Hadoop, Extreme Weather Prediction, Machine Learning, Climate Analytics, MongoDB, Streamlit, Multimodal Data Fusion**

II. Introduction

Extreme rainfall events during the Indian monsoon season frequently cause floods, agricultural loss, and infrastructure damage. Traditional systems face challenges in processing large-scale climate data and generating scalable weather analytics. This creates the need for distributed big data frameworks for efficient rainfall risk prediction.

This project proposes a distributed multimodal extreme weather prediction framework using Hadoop, Apache Spark, MongoDB, machine learning, and a Streamlit dashboard. The system processes large environmental datasets including rainfall, satellite precipitation, and weather parameters, with specific focus on monsoon months (June–September). Apache Spark performs distributed ETL and machine learning operations, while MongoDB and Streamlit provide scalable storage and interactive visualization. The proposed system contributes a scalable climate analytics framework capable of rainfall hotspot detection, geospatial monitoring, and extreme weather risk prediction.

III. Literature Survey

The rapid increase in climate variability, environmental risks, and large-scale sensor data generation has significantly accelerated the adoption of big data analytics and machine learning techniques in environmental monitoring systems. Several recent studies have explored distributed computing frameworks, artificial intelligence models, and multimodal analytics for predictive decision-making across transportation, healthcare, environmental monitoring, and climate systems.

A. Big Data Analytics in Intelligent Systems

Jan et al. [1] presented a comprehensive survey on intelligent transportation systems using big data analytics and artificial intelligence. Their work analyzed distributed architectures, IoT integration, machine learning frameworks, and scalability challenges in intelligent transportation environments. The study emphasized the importance of real-time analytics and multimodal data processing for predictive risk management. However, the work primarily focused on transportation systems and did not address climate-specific prediction pipelines or distributed weather analytics.

Gkikas et al. [2] proposed an Apache Spark-based predictive analytics framework for mortality prediction in Mediterranean aquaculture systems. Their work demonstrated the capability of Spark in processing large-scale environmental datasets for risk prediction applications. The study highlighted distributed processing efficiency but was domain-specific to aquaculture and lacked geospatial climate visualization capabilities.

Patro et al. [3] introduced a remote-sensed intelligent visual analytics framework for environmental and earth monitoring systems. Their work integrated remote sensing datasets with big data analytics for environmental observation and geospatial analysis. The study demonstrated the effectiveness of distributed visual analytics in environmental monitoring but did not incorporate machine learning-based extreme weather classification models.

B. Machine Learning and Risk Prediction Systems

The work in [4] proposed a real-time big data AI engine for enterprise healthcare risk prediction using streaming analytics optimization. The authors employed distributed machine learning pipelines and real-time analytics frameworks for scalable healthcare prediction systems. Although the system demonstrated strong predictive capabilities, the study focused on healthcare analytics rather than environmental forecasting.

Tamrakar and Surendar [5] developed an AI-driven big data framework for climate-resilient transportation infrastructure. Their research emphasized the role of predictive analytics and machine learning in improving infrastructure resilience under changing climatic conditions. The study highlighted climate-aware analytics but lacked multimodal weather fusion and distributed rainfall forecasting pipelines.

Thakur et al. [6] reviewed machine learning techniques for power generation prediction using big data analytics. Their work compared several predictive models and emphasized the importance of scalable distributed processing in handling energy-related big data systems. However, their focus remained on power systems rather than climate and rainfall forecasting.

Cao [7] proposed an AI-enhanced prediction system for emergency department demand during extreme temperature events. The work utilized multiple machine learning models for predictive healthcare analytics under climate-induced emergency scenarios. While the study successfully integrated environmental risk factors, it did not employ distributed weather data fusion or rainfall prediction techniques.

C. Remote Sensing and Environmental Monitoring

Dritsas and Trigka [8] presented a survey on remote sensing and geospatial analysis in the big data era. Their work analyzed recent developments in geospatial analytics, satellite data processing, and large-scale remote sensing frameworks. The study highlighted the growing importance of multimodal geospatial data integration for environmental monitoring systems. However, the research primarily focused on survey analysis rather than predictive climate modeling.

Murugesakumar et al. [9] proposed an air quality monitoring framework using LSTM networks and big data analytics. Their work demonstrated the capability of deep learning architectures in modeling temporal environmental patterns. The study achieved strong predictive performance for air quality forecasting but lacked distributed multimodal weather analytics and rainfall risk visualization systems.

Hassan [10] discussed challenges and opportunities associated with big data and machine learning in autonomous vehicle navigation systems. The study highlighted scalability, sensor fusion, and predictive intelligence within AI-driven systems. Although the research provided insights into multimodal machine learning pipelines, it was unrelated to environmental forecasting or climate analytics.

Nti et al. [11] presented a mini-review of machine learning in big data analytics, focusing on applications, challenges, and future prospects. Their work summarized the integration of machine learning algorithms with distributed big data frameworks across several domains. The study emphasized scalability, predictive modeling, and distributed computation as key research directions for future intelligent systems.

D. Research Gap Analysis

Existing research demonstrates substantial progress in big data analytics, distributed computing, machine learning, and remote sensing applications. However, several limitations remain in current climate prediction systems:

- Most existing studies focus on single-domain applications such as transportation, healthcare, aquaculture, or air quality monitoring.
- Limited research integrates multimodal climate datasets including satellite rainfall data, weather parameters, and reanalysis datasets within a unified distributed prediction framework.
- Many systems lack scalable Hadoop and Apache Spark-based ETL architectures for handling large-scale environmental datasets.
- Existing environmental monitoring frameworks often do not provide interactive NoSQL-based visualization dashboards for real-time climate analytics.
- Few studies combine distributed machine learning pipelines with geospatial risk visualization for extreme rainfall prediction.

To address these limitations, the proposed system introduces a distributed multimodal extreme weather prediction framework using Hadoop, Apache Spark, MongoDB, and machine learning algorithms. The system integrates GPM rainfall data, ERA5 climate reanalysis datasets, and atmospheric weather parameters to predict extreme rainfall events across Indian districts. Additionally, an interactive Streamlit dashboard provides geospatial rainfall risk visualization, hotspot analysis, and climate intelligence insights.

IV. Proposed System Overview

The proposed system presents a distributed multimodal extreme weather risk prediction framework designed for large-

TABLE I
COMPARATIVE ANALYSIS OF EXISTING WORKS

Reference	Domain	Technology Used	Strength	Limitation
Jan et al. [1]	Intelligent Transportation	Big Data + AI	Scalable analytics	No climate prediction
Gkikas et al. [2]	Aquaculture Risk Prediction	Apache Spark	Distributed prediction	Domain-specific
Patro et al. [3]	Environmental Monitoring	Remote Sensing Analytics	Visual analytics	No ML rainfall prediction
IJRPETM [4]	Healthcare Prediction	Streaming AI Analytics	Real-time processing	Not environment-focused
Tamrakar et al. [5]	Climate Infrastructure	AI + Big Data	Climate resilience	No distributed weather fusion
Thakur et al. [6]	Power Generation	ML + Big Data	Comparative ML analysis	No climate analytics
Cao [7]	Emergency Prediction	Multi-model AI	Climate-aware prediction	No rainfall forecasting
Dritsas et al. [8]	Remote Sensing	Geospatial Analytics	Satellite data integration	Survey-based study
Murugesakumar et al. [9]	Air Quality Monitoring	LSTM + Big Data	Temporal modeling	No multimodal rainfall fusion
Hassan [10]	Autonomous Navigation	Big Data + ML	Sensor fusion	Unrelated to weather systems
Nti et al. [11]	Big Data Analytics	ML Frameworks	Broad review	Generalized survey

scale rainfall analytics and climate risk assessment across Indian districts. The framework integrates heterogeneous environmental datasets using Hadoop Distributed File System (HDFS), Apache Spark-based distributed processing, machine learning prediction pipelines, MongoDB-based storage, and an interactive Streamlit analytics dashboard.

The primary objective of the system is to predict extreme rainfall events during the Indian monsoon season using multimodal climate indicators and scalable big data technologies. Unlike conventional rainfall forecasting systems, the proposed framework combines satellite rainfall observations, climate reanalysis datasets, and atmospheric weather parameters into a unified distributed analytics pipeline.

A. Overall Architecture

The system architecture consists of six major layers:

1) Data Collection Layer
2) Distributed Storage Layer
3) ETL and Data Processing Layer
4) Feature Engineering and Label Generation Layer
5) Machine Learning Prediction Layer
6) Visualization and Analytics Dashboard Layer

B. Data Collection Layer

The system utilizes multiple climate-related datasets collected from heterogeneous sources:

- Indian district-wise rainfall dataset
- GPM (Global Precipitation Measurement) satellite rainfall dataset
- ERA5 climate reanalysis dataset
- Weather parameter dataset containing temperature, humidity, and wind speed

These datasets provide multimodal environmental information required for accurate extreme rainfall prediction.

C. Monsoon-Specific Filtering

Since extreme rainfall events in India are predominantly concentrated during the southwest monsoon season, the proposed system specifically focuses on monsoon months:

June, July, August, and September

During the ETL pipeline, all datasets were filtered to retain only monsoon-season records. This filtering significantly improved data consistency, reduced irrelevant dry-season observations, minimized missing value propagation, and enhanced model learning performance for rainfall-related climate patterns.

The monsoon filtering strategy also reduced computational overhead in distributed Spark processing while improving prediction relevance for real-world extreme weather analysis.

D. Distributed Storage and Processing

All datasets were stored in Hadoop Distributed File System (HDFS) for scalable distributed storage. Apache Spark was used for parallel ETL operations including:

- Data cleaning
- District normalization
- Date standardization
- Missing value handling
- Dataset joining
- Aggregation and duplicate removal

Spark DataFrames and distributed transformations enabled efficient handling of large-scale environmental datasets.

E. Data Cleaning and Preprocessing

The collected multimodal climate datasets contained inconsistent district names, duplicate records, missing values, and date formatting issues. To improve data quality and ensure reliable distributed processing, a comprehensive preprocessing

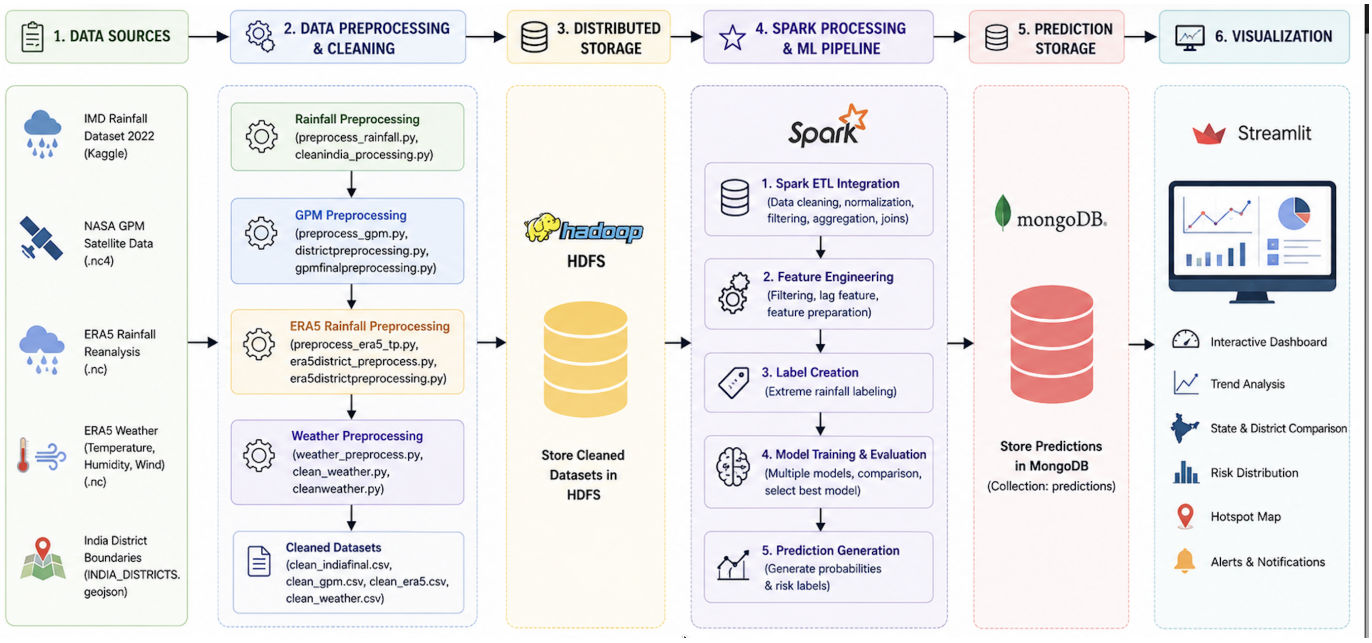

Fig. 1. Architecture of the proposed distributed extreme rainfall prediction framework.

pipeline was implemented using both Pandas and Apache Spark.

Initially, dataset columns were standardized by converting all column names to lowercase. District names were normalized by converting them to uppercase, removing special characters, eliminating extra spaces, and correcting inconsistent naming patterns across datasets. Several district mappings such as `BANGALORE` → `BENGALURU` and `TRIVANDRUM` → `THIRUVANANTHAPURAM` were manually standardized to improve join consistency.

Date columns were converted into proper datetime formats, and invalid or null records were removed. Duplicate environmental observations generated from multiple weather entries were resolved using aggregation operations. Apache Spark group-based aggregation was used to compute average rainfall and weather parameters for each district-date combination.

To improve climate relevance and reduce seasonal noise, the datasets were filtered to retain only Indian monsoon months (June–September), where most extreme rainfall events occur. After preprocessing, the cleaned datasets were stored in Hadoop Distributed File System (HDFS) and used for distributed ETL processing, feature engineering, and machine learning model training.

F. Feature Engineering

Feature engineering was performed using Apache Spark to generate meaningful climate features for machine learning-based rainfall risk prediction. Initially, rows containing no rainfall signal were removed by filtering records where both GPM rainfall and ERA5 rainfall values were zero. Invalid weather observations with zero temperature or humidity values were also eliminated to improve dataset reliability.

A temporal rainfall feature named `rainfall_lag1` was created using Spark window functions. This feature captures the previous day's rainfall for each district and helps the model learn rainfall continuity and temporal weather patterns.

The final feature set included:

- GPM rainfall
- ERA5 rainfall
- Temperature
- Humidity
- Wind speed
- Previous day rainfall (`rainfall_lag1`)

Apache Spark DataFrame operations and distributed window processing enabled scalable feature generation across large climate datasets stored in HDFS.

G. Label Creation

To perform supervised machine learning for extreme rainfall prediction, binary class labels were generated from the target rainfall values. Apache Spark was used to calculate rainfall distribution statistics efficiently on the distributed dataset stored in HDFS.

A percentile-based thresholding approach was adopted to identify extreme rainfall events. The 95th percentile of the `target_rainfall` distribution was selected as the threshold for classification. Rainfall values greater than or equal to this threshold were labeled as extreme rainfall events.

The binary label definition was:

- 1 → Extreme Rainfall Event
- 0 → Normal Rainfall Event

This approach enabled automatic identification of high-risk rainfall events while handling large-scale climate data efficiently using distributed Spark operations. The labeled

dataset was then stored in HDFS for machine learning model training and evaluation.

H. Machine Learning Prediction Pipeline

Several machine learning algorithms were trained and evaluated using Apache Spark MLlib:

- Logistic Regression
- Random Forest Classifier
- Decision Tree Classifier
- Gradient Boosted Trees (GBT)

```
🚀 Spark Started
📊 Data Loaded
✅ Data Cleaned
⚖️Class weights added
Train: 15339 | Test: 3763

🔷 Training Logistic Regression
✅ Logistic Regression AUC = 0.7368

🔷 Training Random Forest
Java HotSpot(TM) 64-Bit Server VM warning: CodeCache is full. Compiler has be
Java HotSpot(TM) 64-Bit Server VM warning: Try increasing the code cache size
CodeCache: size=131072Kb used=43777Kb max_used=43782Kb free=87294Kb
 bounds [0x0000000106bcc000, 0x00000001096dc000, 0x000000010ebcc000]
 total_blobs=15477 nmethods=14203 adapters=1184
 compilation: disabled (not enough contiguous free space left)
✅ Random Forest AUC = 0.7753

🔷 Training Decision Tree
✅ Decision Tree AUC = 0.3158

🔷 Training GBT
✅ GBT AUC = 0.7748

🏆 BEST MODEL: Random Forest
📈 BEST AUC: 0.7753224937022303
💾 Model Saved
```

Fig. 2. Distributed machine learning model training and performance evaluation using Apache Spark

Figure 2 illustrates the distributed training and evaluation of multiple machine learning models using Apache Spark MLlib. Among the evaluated classifiers, the Random Forest model achieved the best prediction performance with an AUC score of approximately 0.775 for extreme rainfall risk classification. Class imbalance handling was performed using weighted class learning techniques.

The dataset was divided into training and testing subsets using an 80:20 split ratio. Model evaluation was conducted using Area Under ROC Curve (AUC) metrics.

Among all models, Random Forest achieved the best predictive performance with the highest AUC score and was selected as the final prediction model.

I. Prediction Storage using MongoDB

Prediction outputs generated by the trained machine learning model were stored in MongoDB, a NoSQL document database. MongoDB was selected because of its flexibility in handling semi-structured prediction data and real-time analytics requirements.

```
✅ Model Loaded
⚡ Predictions Generated: 19102
+------------------+------------------+-------------+----------+
|gpm_rainfall      |era5_rainfall     |rainfall_lag1|prediction|
+------------------+------------------+-------------+----------+
|0.197763149368421 |0.0               |0.0          |0.0       |
|30.044736315789475|0.0               |0.0          |1.0       |
|0.0               |6.6757202E-4      |10.7175      |0.0       |
|0.0               |0.00638961794     |0.00160933   |0.0       |
|0.0               |0.458526612       |0.0          |0.0       |
|0.0               |4.45327762        |0.0          |0.0       |
|0.0               |1.63955688        |0.106501     |0.0       |
|2.9192855397142856|0.490188594       |0.0          |0.0       |
|38.44789415789474 |0.490379332       |0.0          |1.0       |
|0.0               |0.5261421260000001|29.5045      |1.0       |
+------------------+------------------+-------------+----------+
only showing top 10 rows

+----------+--------+---------+-----------------------+----------+
|      date|district|    state|extreme_rain_probability|risk_label|
+----------+--------+---------+-----------------------+----------+
|2022-06-06|ADILABAD|Telangana|     0.31743407616891595|       0.0|
|2022-06-07|ADILABAD|Telangana|      0.6302797915216757|       1.0|
|2022-06-12|ADILABAD|Telangana|      0.19383643111142068|       0.0|
|2022-07-01|ADILABAD|Telangana|     0.21284556301271018|       0.0|
|2022-07-03|ADILABAD|Telangana|     0.13669456725459544|       0.0|
|2022-07-04|ADILABAD|Telangana|    0.054797735619417164|       0.0|
|2022-07-05|ADILABAD|Telangana|     0.20844182935581496|       0.0|
|2022-07-06|ADILABAD|Telangana|     0.07895386518454044|       0.0|
|2022-07-07|ADILABAD|Telangana|      0.8996492938190545|       1.0|
|2022-07-08|ADILABAD|Telangana|      0.6103566965441843|       1.0|
+----------+--------+---------+-----------------------+----------+
only showing top 10 rows
```

Fig. 3. Extreme rainfall prediction generation using the trained Random Forest model

Figure 3 presents the prediction output generated using the trained Random Forest classifier on the processed multimodal rainfall dataset. The system predicts rainfall risk probabilities and classifies extreme rainfall events using satellite rainfall indicators and temporal lag-based features. The stored prediction fields included:

- District
- State
- Date
- Extreme rainfall probability
- Predicted risk label

J. Interactive Climate Analytics Dashboard

An interactive climate analytics dashboard was developed using Streamlit to provide real-time visualization, monitoring, and analytical insights for extreme rainfall prediction results. The dashboard integrates machine learning predictions stored in MongoDB and enables dynamic exploration of rainfall risk patterns across multiple Indian states and districts.

The dashboard was designed as a scalable visualization layer on top of the distributed Hadoop–Spark architecture, enabling efficient analysis of large-scale climate datasets generated during prediction workflows.

1) Dashboard Features: The dashboard provides multiple analytical and interactive components for rainfall risk monitoring:

- Real-time prediction visualization using MongoDB-stored prediction outputs
- State-wise and district-wise rainfall risk analysis
- Interactive temporal trend visualization of rainfall probability
- Extreme rainfall hotspot identification using geospatial maps

- Probability distribution analysis for extreme rainfall events
- Heatmaps for state-wise extreme event density
- Daily and monthly rainfall trend analytics
- Automated high-risk rainfall alert generation
- Machine learning model performance visualization
- Big data system architecture overview and AI-based insights

2) Geospatial Rainfall Risk Mapping: A geospatial rainfall hotspot visualization module was implemented using Plotly Mapbox integration. District-level latitude and longitude coordinates were mapped to prediction results for identifying high-risk rainfall zones across India.

Extreme rainfall districts predicted by the Random Forest model were visualized dynamically using interactive marker-based maps. Marker size and color intensity represented the predicted extreme rainfall probability, allowing easy identification of vulnerable regions during monsoon months.

3) Analytical Insights and Monitoring: The dashboard includes several statistical and analytical visualizations for understanding rainfall behavior and prediction trends:

- Time-series analysis of rainfall probability trends
- Top high-risk district ranking visualization
- Extreme versus normal rainfall event distribution
- Monthly monsoon risk pattern analysis
- State-level rainfall density heatmaps
- Daily extreme rainfall event frequency monitoring

The dashboard also provides AI-generated analytical summaries highlighting:

- Highest-risk states and districts
- Maximum and average rainfall probabilities
- Total predicted extreme rainfall events
- Monsoon season dominance in rainfall occurrence
- Impact of lag rainfall features on prediction quality

4) Technology Stack: The visualization framework was developed using the following technologies:

- Streamlit for interactive dashboard development
- Plotly for advanced interactive visualizations
- MongoDB for storing prediction outputs
- Apache Spark for distributed prediction generation
- Hadoop HDFS for scalable data storage

The dashboard acts as the final analytical layer of the proposed distributed extreme rainfall prediction framework, enabling real-time decision support and climate risk monitoring.

V. Results and Analysis

A. Evaluation Metrics

The performance of the machine learning models was evaluated using the Area Under Receiver Operating Characteristic Curve (AUC-ROC). AUC measures the capability of the classifier to distinguish between extreme and non-extreme rainfall events. A higher AUC value indicates better classification performance and robustness.

The evaluation process was carried out on the test dataset generated using an 80:20 train-test split strategy. Class imbalance was handled using weighted learning during model training.

B. Model Performance Comparison

Multiple machine learning models were trained and evaluated for extreme rainfall prediction. The obtained AUC scores are presented in Table II.

TABLE II
Model Performance Comparison

Model	AUC Score
Logistic Regression	0.7368
Random Forest	0.7753
Decision Tree	0.3158
GBT Classifier	0.7748

Among all evaluated models, the Random Forest classifier achieved the highest AUC score of approximately 0.775, indicating superior capability in identifying extreme rainfall events from multimodal climatic features.

C. Prediction Insights

The developed prediction framework generated several meaningful insights regarding extreme rainfall behavior during Indian monsoon seasons.

- Extreme rainfall occurrences were predominantly concentrated during the monsoon months of June to September.
- States such as Kerala, Goa, Maharashtra, and parts of the North-East region exhibited comparatively higher rainfall risk probabilities.
- District-level hotspot visualization revealed clusters of high-risk rainfall zones across coastal and high-altitude regions.
- Satellite-derived rainfall indicators from GPM and ERA5 significantly contributed to improving prediction robustness.
- Incorporating previous-day rainfall as a lag feature enhanced temporal dependency learning and improved model accuracy.
- The probability distribution analysis showed that extreme rainfall events represented a smaller fraction of the total observations, validating the imbalanced nature of the dataset.
- The Streamlit dashboard enabled interactive monitoring of rainfall probabilities, hotspot detection, state-wise risk analysis, and real-time exploration of prediction outputs.

VI. Dashboard Visualization

A. Streamlit Dashboard

Figure 4 presents the main Streamlit dashboard interface developed for real-time monitoring of extreme rainfall predictions. The dashboard integrates interactive filtering, statistical summaries, and temporal rainfall risk visualizations for efficient climate analytics.

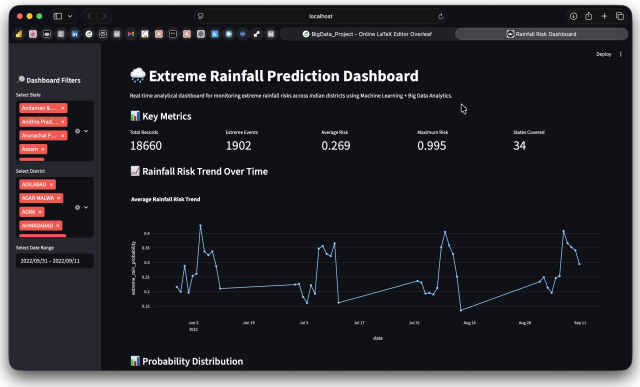

Fig. 4. Main dashboard interface

Figure 5 illustrates the probability distribution of predicted rainfall risks along with comparative state-wise rainfall risk analysis. The visualization helps identify regions exhibiting higher average extreme rainfall probabilities during monsoon months.

Figure 6 shows the geospatial hotspot mapping of predicted extreme rainfall events across Indian districts. Interactive map-based visualization enables easy identification of vulnerable high-risk regions using district-level prediction probabilities.

Figure 7 demonstrates the AI-generated analytical insights and rainfall event distribution obtained from the prediction framework. The dashboard summarizes key statistics such as total extreme events, high-risk states, and model performance observations.

Fig. 5. Probability distribution and state-wise rainfall risk

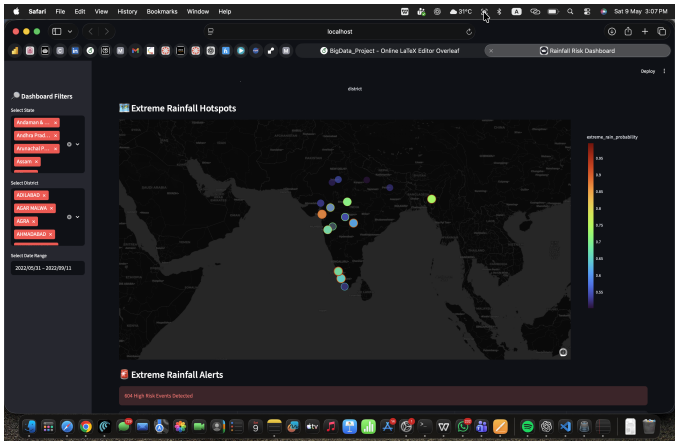

Fig. 6. Geospatial visualization of extreme rainfall

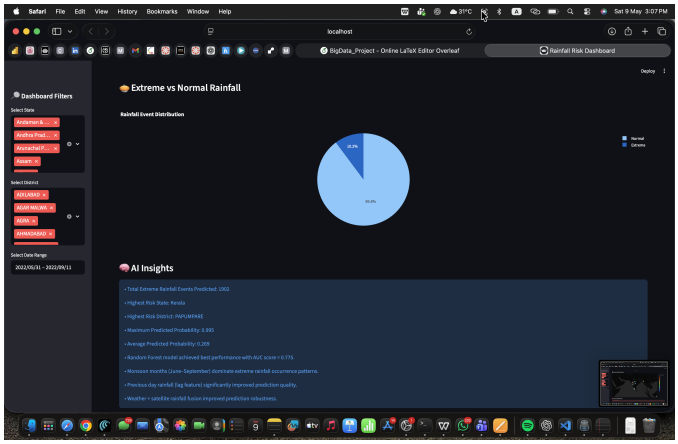

Fig. 7. AI insights and rainfall event distribution

B. Interactive Visualizations

Figure 8 presents comparative analysis of state-wise rainfall risk and top high-risk districts identified by the Random Forest prediction model. The visualization supports regional risk assessment and decision-making processes.

Figure 9 illustrates interactive geospatial visualization of rainfall hotspots using Plotly Mapbox integration. Marker size and color intensity represent predicted rainfall severity across multiple Indian districts.

VII. Advantages of Proposed System

- Distributed climate data processing
- Scalable architecture
- Interactive analytics dashboard
- Multimodal weather fusion
- Efficient prediction pipeline
- Real-time monitoring capability

VIII. Limitations

- Limited geographical coordinate coverage
- Seasonal filtering constraints
- Class imbalance challenges
- Dependence on historical datasets

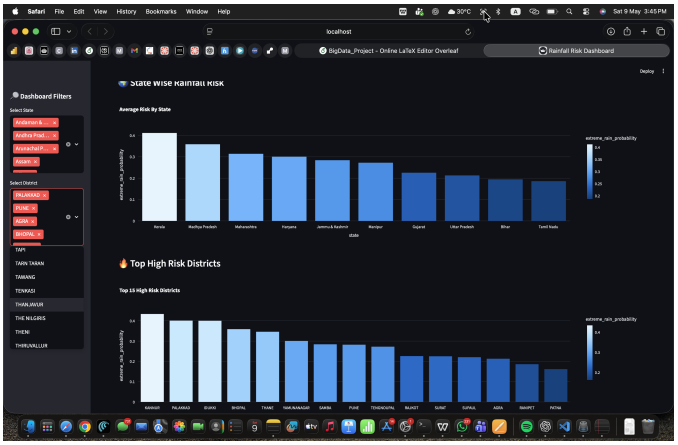

Fig. 8. State-wise rainfall risk and top high-risk district analysis

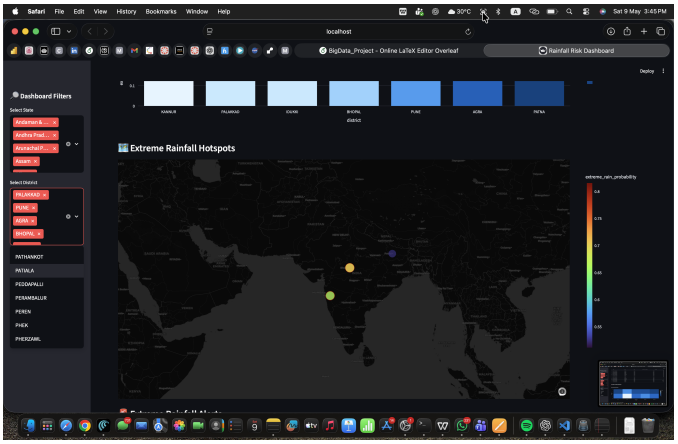

Fig. 9. Extreme rainfall hotspot visualization using geospatial mapping

IX. Future Work

- Deep learning integration
- Real-time streaming analytics
- IoT sensor integration
- Flood prediction systems
- Cloud deployment
- GIS-based advanced visualization

X. Conclusion

This work presented a distributed multimodal extreme rainfall prediction framework using Hadoop, Apache Spark, MongoDB, and Streamlit for scalable climate risk analytics. Multiple climatic and satellite-derived rainfall datasets were integrated and processed using distributed ETL pipelines and Spark-based feature engineering techniques.

Machine learning models were trained to classify extreme rainfall events during Indian monsoon months, with the Random Forest classifier achieving the best prediction performance with an AUC score of approximately 0.775. The inclusion of multimodal rainfall indicators and temporal lag-based features improved prediction robustness and risk identification capability.

An interactive Streamlit dashboard was developed for real-time visualization of rainfall risks, hotspot mapping, state-wise analysis, and AI-generated insights. The proposed framework demonstrates the effectiveness of big data technologies and distributed machine learning systems for scalable environmental monitoring and climate risk prediction applications.

References

[1] M. A. Jan, M. Adil, B. Brik, S. Harous and S. Abbas, "Making Sense of Big Data in Intelligent Transportation Systems: Current Trends, Challenges and Future Directions," *ACM Computing Surveys*, vol. 57, no. 8, Article 197, Aug. 2025. doi: 10.1145/3716371

[2] M. C. Gkikas, D. C. Gkikas, G. Vonitsanos, J. A. Theodorou and S. Sioutas, "Application of Machine Learning for Predictive Analysis and Management of Mediterranean-Farmed Fish Mortalities: A Risk Management Case Study Using Apache Spark," *Applied Sciences*, vol. 14, no. 22, 2024. doi: 10.3390/app142210112

[3] P. Patro et al., "Big Data-Based Remote-Sensed Intelligent Visual Analytics for Environmental and Earth Monitoring Systems," *Journal of Applied Science and Technology Trends*, vol. 6, no. 5, pp. 35–50, 2025. doi: 10.38094/jastt605589

[4] "Real Time Big Data AI Engine for Enterprise Healthcare Risk Prediction with Streaming Analytics Optimization," *International Journal of Research Publications in Engineering, Technology and Management*, vol. 9, no. 1, pp. 154–163, 2026. doi: 10.15662/IJRPETM.2026.0901020

[5] G. Tamrakar and A. Surendar, "AI-Driven Big Data Framework for Climate-Resilient Transportation Infrastructure," *Journal of Smart Infrastructure and Environmental Sustainability*, vol. 2, no. 1, pp. 38–44, 2025. doi: 10.17051/JSIES/02.01.05

[6] A. Thakur et al., "Prediction of Power Generation Using Machine Learning With Big Data Analytics: A Review," in *2025 7th International Symposium on Advanced Electrical and Communication Technologies (ISAECT)*, Mohali, India, 2025. doi: 10.1109/ISAECT68904.2025.11318739

[7] R. Cao, "AI-Enhanced Prediction System for Emergency Department Demand during Extreme Temperature Events: A Multi-Model Approach," in *2025 International Conference on Big Data and Intelligent Medicine (BDIMed)*, Wuhan, China, 2025. doi: 10.1109/BDIMed67722.2025.11309338

[8] E. Dritsas and M. Trigka, "Remote Sensing and Geospatial Analysis in the Big Data Era: A Survey," *Remote Sensing*, vol. 17, no. 3, 2025. doi: 10.3390/rs17030550

[9] B. Murugesakumar et al., "Big Data Analytics with LSTM Networks for Predictive Air Quality Monitoring in Urban Environments," in *2025 Tenth International Conference on Science Technology Engineering and Mathematics (ICONSTEM)*, Chennai, India, 2025. doi: 10.1109/ICONSTEM65670.2025.11374801

[10] N. A. Hassan, "Big Data and Machine Learning in Autonomous Vehicle Navigation: Challenges and Opportunities," *JACAIDMS*, vol. 14, no. 12, pp. 54–64, 2024.

[11] I. K. Nti, J. A. Quarcoo, J. Aning and G. K. Fosu, "A Mini-Review of Machine Learning in Big Data Analytics: Applications, Challenges, and Prospects," *Big Data Mining and Analytics*, vol. 5, no. 2, pp. 81–97, 2022. doi: 10.26599/BDMA.2021.9020028